BMC Oral Health

SYSTEMATIC REVIEW Open Access

Umbrella review of photobiomodulation/low-level laser therapy and vitamin D in orthodontics: effects on tooth movement, pain, root resorption, and bone biomarkers

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Abstract

Background To synthesize evidence from systematic reviews and meta-analyses on low-level laser therapy/Photobiomodulation (LLLT/PBM) and vitamin D as non-invasive adjuncts for accelerating orthodontic tooth movement (OTM) and mitigating associated side effects.

Methods PubMed/MEDLINE, Embase, Scopus, Web of Science, and Cochrane Library databases were searched from inception to November 18, 2025. SRMAs primarily comprising human RCTs evaluating LLLT/PBM or vitamin D in orthodontics were included; in some mixed-design reviews, clearly separable human orthodontic data from a small number of CCTs were retained narratively. (one pre-registered exception with 4 RCTs for vitamin D to ensure completeness). Randomized controlled trials were prioritized; some retained reviews also included a small number of controlled clinical trials when relevant human orthodontic data were clearly separable. Quality was assessed using AMSTAR-2, ROBIS, and GRADE. Narrative synthesis was performed because of high heterogeneity.

Results Twenty systematic reviews and meta-analyses covering a large number of human orthodontic trials were included. Low-certainty evidence suggests that LLLT/PBM may be associated with accelerated OTM, slight reductions in orthodontically induced inflammatory root resorption (OIIRR), and short-term pain relief. Vitamin D showed inconsistent findings across outcomes with no reliable overall clinical benefit.

Conclusions Current evidence is insufficient to recommend the routine clinical use of LLLT/PBM or vitamin D in orthodontics.

Trial Registration PROSPERO CRD42024591253.

Plain Language summary

Orthodontic treatment often involves long treatment durations, potentially causing discomfort, root damage, and reduced compliance. This review examines evidence from systematic reviews and meta-analyses on low-level laser therapy/photobiomodulation (LLLT/PBM) and vitamin D as possible aids for faster tooth movement through bone remodeling. Low-certainty evidence suggests that LLLT/PBM may be associated with slightly faster

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tooth movement, minor reductions in root resorption, and pain relief. However, the high variability among studies reduces confidence in these suggestions. Vitamin D showed no consistent association with accelerated OTM or other outcomes (very low certainty evidence); exploratory subgroup findings on local administration were inconsistent and insufficient for recommendations. Overall, inconsistencies limit reliable conclusions and highlight the need for better-designed trials to provide clearer guidance on this topic.

Keywords Photobiomodulation, Low-level laser therapy, Vitamin D, Orthodontic tooth movement, Root resorption, Umbrella review

Introduction

Orthodontic treatment with fixed appliances typically lasts between 18 and 24 months [1]. Treatment may also be accompanied by adverse effects such as orthodontically induced inflammatory root resorption (OIIRR) and pain [2–4]. These burdens can negatively influence the patient experience and the perceived value of treatment [5, 6].

Non-surgical adjuncts such as low-level laser therapy/photobiomodulation (LLLT/PBM) and vitamin D supplementation have been explored for their potential to influence orthodontic tooth movement (OTM) [7–18]. LLLT/PBM uses red-to-near-infrared light, and orthodontic applications have typically involved wavelengths within the broader therapeutic laser range used in dentistry [7–11, 19–23]. PBM has been proposed to modulate biologic processes involved in bone remodeling, including the RANK/RANKL/OPG pathway [24]. Vitamin D (calcitriol) activates the vitamin D receptor (VDR) in periodontal tissues and may influence bone remodeling and OTM through effects on the RANKL/OPG pathway [15, 24, 25].

However, published systematic reviews and meta-analyses have reported inconsistent findings, likely owing to variations in dosimetry, application methods, follow-up intervals, and risk of bias [7–15, 26]. Mechanistic overlaps between PBM-related signaling and vitamin D pathways have led to hypotheses about possible synergy, but SRMAs evaluating combined LLLT/PBM and vitamin D interventions are lacking. Genetic factors, including VDR polymorphisms, may also influence individual responses and present opportunities for more personalized orthodontic care [25]. An umbrella review is therefore warranted to resolve discrepancies, assess evidence certainty using established appraisal frameworks [27–30], and guide clinical practice. Patient-centered outcomes and cost-effectiveness are also important when judging the practical value of adjunctive approaches in orthodontics [5, 6]. This review synthesizes human SRMAs on OTM (primary), OIIRR, pain, and biomarkers (secondary), identifies knowledge gaps, and highlights unresolved issues related to combined interventions and personalized approaches based on genetic variability.

Methods

The protocol was registered with PROSPERO (CRD42024591253). Biomarkers were prespecified as secondary outcomes, and any meta-regression findings reported within the included SRMAs, including the vitamin D administration route (local vs. systemic), were extracted narratively where available. The review was conducted and reported in accordance with PRISMA 2020 [31] and broader contemporary evidence-synthesis standards [32]. Methodological appraisal was based on AMSTAR-2 [27], ROBIS [30], GRADE [28], and umbrella-review guidance [29].

PICO question

Population

Orthodontic patients aged ≥ 12 years undergoing fixed-appliance treatment in human clinical studies, with randomized controlled trials prioritized.

Intervention

Application of LLLT/PBM (GaAlAs diodes or LEDs at < 100 mW/cm^2, 600–1200 nm wavelengths) or vitamin D (oral/systemic doses of 1–50 µg/day or local PDL/gingival injections of 0.1–1 µg calcitriol).

Comparison

Sham procedures, placebo, or standard orthodontic care without adjuncts.

Outcome

Impact of LLLT/PBM or vitamin D on OTM acceleration (primary) and related outcomes (secondary: OIIRR, pain, and biomarkers such as RANKL/OPG in GCF/serum).

Focused questions

What is the impact of LLLT/PBM and vitamin D supplementation on orthodontic tooth movement in patients undergoing fixed-appliance orthodontic treatment?

What patterns emerged regarding the LLLT/PBM and vitamin D application parameters across the included SRMAs?

Inclusion criteria

The SRMAs included in this study synthesized data from human randomized controlled trials (RCTs; parallel or

quasi-randomized designs) evaluating LLLT/PBM or vitamin D as adjuncts to fixed orthodontic therapy. RCTs were prioritized; some retained reviews also included a small number of controlled clinical trials (CCTs) when relevant human orthodontic data were clearly identified and could be synthesized narratively. Because the evidence base for vitamin D was limited, we exceptionally retained one review with four primary RCTs [14] per protocol; all other included SRMAs met the ≥ 5 primary-study threshold. Previous umbrella and scoping reviews were excluded from the formal synthesis to maintain methodological consistency and prevent overlap.

Exclusion criteria

We excluded animals or in vitro-only SRMAs, those with fewer than five studies, non-English SRMAs, duplicates, and those with surgical confounders to prevent overlap (Table 1).

Search strategy

A literature search was conducted across PubMed/MEDLINE, Embase, Scopus, Web of Science Core Collection, and Cochrane Library from their inception to November 18, 2025. The search combined MeSH/Emtree terms and keywords: ("orthodontic tooth movement" OR "OTM" OR "tooth displacement") AND [("low-level laser therapy" OR "LLLT" OR "photobiomodulation" OR "PBM" OR "light-emitting diode") OR ("vitamin D" OR "calcitriol" OR "cholecalciferol" OR "1,25-dihydroxyvitamin D")] AND ("systematic review" OR "meta-analysis" OR "overview"). The search identified 2,047 records, which were reduced to 1,723 after deduplication using EndNote X9 software. The gray literature search across ClinicalTrials.gov (n = 45 records), WHO International Clinical Trials Registry Platform ($n = 28$), OpenGrey ($n = 12$), ProQuest Dissertations & Theses ($n = 35$), and Google Scholar ($n = 150$, first 10 pages) yielded no additional eligible SRMAs after deduplication and screening.

Table 1 List of excluded studies

No	Author and Year	Journal	Intervention/topic	Reason for Exclusion
1	Sousa MV et al. 2014 [33]	Photomedicine and Laser Surgery	Low-level laser therapy / photobiomodulation	Excluded because it is an older systematic review without meta-analysis, combines orthodontic tooth movement with pain-control outcomes, and does not provide a focused SRMA dataset aligned with the umbrella-review eligibility criteria.
2	Nimeri G et al. 2013 [34]	Progress in Orthodontics	General acceleration methods	Excluded because it is a narrative overview rather than a reproducible systematic review/meta-analysis, with no formal risk-of-bias appraisal or quantitative synthesis.
3	Kale S et al. 2004 [35]	American Journal of Orthodontics and Dentofacial Orthopedics	Vitamin D / prostaglandin E2	Excluded because it is a primary animal study, not a systematic review/meta-analysis, and does not meet the human-RCT review-level eligibility criterion.
4	Al-Khalifa KS et al. 2021 [36]	European Journal of Dentistry	Micro-osteoperforations	Excluded because the intervention is micro-osteoperforation, a minimally invasive surgical acceleration method, outside the predefined LLLT/PBM and vitamin D scope and a potential surgical confounder.
5	Long H et al. 2013 [26]	The Angle Orthodontist	Interventions for accelerating orthodontic tooth movement	Excluded because it is a broad systematic review of multiple acceleration methods (including surgical interventions) and does not focus specifically on LLLT/PBM or vitamin D as standalone interventions.
6	Yavagal et al. 2021 [37]	International Journal of Clinical Pediatric Dentistry	Laser photobiomodulation	Excluded because the review was pediatric-focused and substantially overlapped with broader PBM reviews; retained as contextual/supportive evidence if pediatric PBM evidence is discussed separately.
7	Kalemaj Z et al. 2015 [17]	European Journal of Oral Implantology	Surgical and non-surgical acceleration methods	Excluded because it mixes surgical and non-surgical interventions, includes surgical confounders, and the LLLT-only subset is below the protocol threshold for a focused PBM synthesis.
8	Almoammar K (2018) [38]	Clin Cosmet Investig Dent	Vitamin D / narrative insight review	Excluded because it is a narrative/insight review rather than a systematic review or meta-analysis of human RCTs. It lacks a reproducible search strategy, deduplication process, formal risk-of-bias appraisal, and quantitative synthesis meeting the umbrella-review eligibility criteria.
9	Jiménez-Peña OM et al. (2025) [39]	Evid Based Dent	Photobiomodulation / LLLT for orthodontic tooth movement (umbrella review)	Excluded because it is itself an umbrella review (overview of systematic reviews) rather than a primary SRMA synthesizing human orthodontic RCTs directly. Previous umbrella and scoping reviews were excluded per protocol to maintain methodological consistency and prevent overlap.
10	Zamanian MY et al. (2024) [40]	Food Sci Nutr	Vitamin D3 / narrative review on orthodontic tooth movement	Excluded because it is explicitly structured and labeled as a narrative review. It does not meet the requirements for a systematic review/meta-analysis (no protocol, limited formal search/deduplication, no risk-of-bias assessment or meta-analysis of ≥5 human RCTs).

The complete search string is provided in Supplementary File 2.

Selection of studies and data extraction

A PRISMA-recommended flow diagram was used for the literature search [31]. As shown in Fig. 1 (PRISMA 2020 flow diagram for study selection, illustrating 2,047 initial records, 1,723 after deduplication, 116 full-text assessed, and 20 included SRMAs), two reviewers (SAAB and AHR) screened the titles and abstracts before conducting a full-text review to apply the inclusion and exclusion criteria. Conflicts were resolved through consultation with a third reviewer (RH). Any meta-regression findings reported by the included SRMAs were extracted, where available, to describe potential moderators, such as dosimetry, age, and sex. For SRMAs with multi-arm trials or multiple time points, we extracted the most relevant human data (primary endpoint at 1–3 months of follow-up). Data were extracted using a standardized data extraction form (Supplementary File 3).

General characteristics of the included studies

The general characteristics of the included studies were extracted, including the author, title, journal, year, and SCImago Journal Rank quartile. The features included article type, number of RCTs, databases searched, objectives, conclusions, and study limitations. All data below were extracted directly from the source SRMAs as originally reported; no new primary data were collected and no independent analyses were performed by the authors. Human data originated from mixed populations across the source reviews. Dual independent data extraction showed < 4% disagreement, resolved by consensus. In Table 2, the final column summarizes the conclusions reported by each source SRMA; these statements are presented descriptively.

Statistical analysis

Effect sizes, confidence intervals, and heterogeneity statistics were extracted directly from the included SRMAs as originally reported. No de novo second-order meta-analysis or formal statistical synthesis was performed. Our role was limited to narrative synthesis, quality appraisal (AMSTAR-2, ROBIS, GRADE), and a descriptive summary of the findings from the source reviews. Publication bias, study overlap, and sensitivity analyses mentioned in the source SRMAs were extracted and

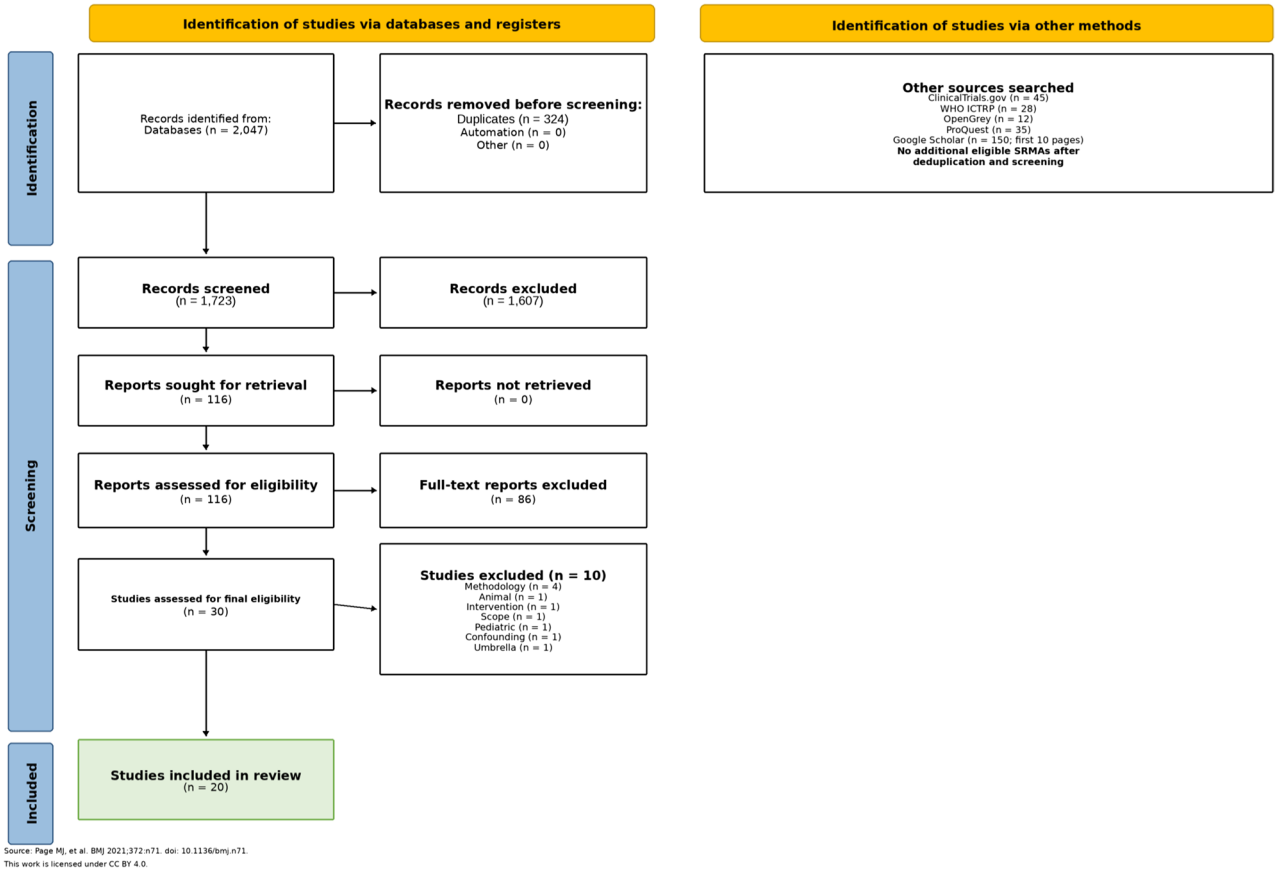


Fig. 1 PRISMA 2020 flow diagram for study selection

Table 2 Characteristic findings from the 20 Included SRMAs

No	Author (Year)	Type of review	No. of primary studies (design)	Population (patients; age range, % female)	Intervention (human details)	Key human outcomes (pooled effects where available)	Heterogeneity (reported as I^2 or narrative summary)	AMSTAR-2 overall confidence	GRADE certainty (human data)	Main conclusion of the review presented in the SRMA
1	Domínguez Camacho (2020) [7]	Systematic review	9 RCTs	260; 15–27 y, 58%	PBM: 780–830 nm, 4–10 J/cm^2, 3×/wk	OTM + 24% speed (qualitative average)	NR; dosimetry	Moderate	Low	Effective wavelength range identified; acceleration possible
2	Jedliński (2020) [8]	SR + meta-analysis	6 RCTs, 2 CCTs	220; 14–26 y, 60%	PBM intraoral: 660–850 nm, 5–8 J/cm^2	Treatment time ↓ ~ 25%; est. MD = 0.45 mm/month; intraoral superior	35%; route	High	Moderate	Significant acceleration, intraoral devices superior
3	Ge (2014)	SR + meta-analysis	6 RCTs, 3 quasi-RCTs	211; 12–25 y, 55%	LLLT: 5–25 J/cm^2, 810 nm GaAlAs	OTM MD = 0.19 mm (7 d) to 1.08 mm (2 mo)	45%; density	Moderate	Low	LLLT accelerates OTM; low dose more effective
4	Bakdach & Hadad (2020) [10]	SR + meta-analysis	25 RCTs	680; 13–29 y, 65%	LLLT canines: 2.5–25 J/cm^2, 660–980 nm	Upper canine MD 0.49–0.50 mm, lower 0.28–0.52 mm (2–3 mo)	90%; dose, phase	Critically Low	Very low	Acceleration confirmed but high heterogeneity limits confidence
5	de Almeida (2016) [11]	SR + meta-analysis	5 RCTs, 1 quasi-RCT	150; 14–28 y, 62%	LLLT extraction: 2–8 J/cm^2, 780 nm	Maxilla MD = 0.45 mm (3 mo), mandible 0.32 mm (1 mo); overall NS	72%; jaw	Moderate	Low	Limited evidence of acceleration, maxillary more responsive
6	Deana (2017) [2]	SR + meta-analysis (pain)	20 RCTs	540; 12–24 y, 59%	LLLT pain: 780–904 nm, 4–10 J/cm^2	Spontaneous SMD = −0.68 (24 h), chewing SMD = −0.72 (72 h)	42%; timing	Moderate	Low	Effective short-term pain reduction
7	Imani (2018) [41]	SR + meta-analysis	6 RCTs	180; 14–25 y, 57%	LLLT canines: 810 nm, 8 J/cm^2	MD 0.40–1.53 mm over 4.5 months	28%; duration	High	Moderate	Strong evidence of canine acceleration
8	El-Angbawi (2023) [42]	Cochrane review (LLLT subset)	12 RCTs	520; 12–34 y, 61%	PBM adjuncts < 10 J/cm^2	~ 0.35 mm/month additional movement	75%; adjunct type	High	Low	Weak evidence of acceleration
9	Zheng (2023) [43]	SR + meta-analysis	8 RCTs	240; 13–30 y, 64%	LLLT alignment: 810 nm, 5 J/cm^2	Levelling time ↓ mean 30.36 days	27%; archwire	High	Moderate	Significant reduction in alignment duration
10	Huang (2023) [44]	SR + meta-analysis	5 RCTs, 3 CCTs	280; 12–28 y, 60%	PBM alignment LEDs 625–850 nm	Rate + 25%, duration −22%	65%; device	Moderate	Low	Moderate acceleration in alignment phase

Table 2 (continued)

No	Author (Year)	Type of review	No. of primary studies (design)	Population (patients; age range, % female)	Intervention (human details)	Key human outcomes (pooled effects where available)	Heterogeneity (reported as I^2 or narrative summary)	AMSTAR-2 overall confidence	GRADE certainty (human data)	Main conclusion of the review presented in the SRMA
11	AlShahrania (2019)	SR + meta-analysis	10 RCTs	320; 12–22 y, 56%	PBM GaAlAs 780 nm, 6 J/cm^2	MD = 0.59 mm additional movement	95%; settings	Moderate	Low	Acceleration observed but very high heterogeneity
12	Cronshaw (2019) [20]	Systematic review	15 RCTs	400; 12–30 y, 55%	Various PBM wavelengths 4–10 J/cm^2	Qualitative acceleration & pain reduction	70%; dosimetry, phase	Moderate	Low	Reported support for PBM use, while acknowledging limitations
13	Elgadi (2023) [21]	Systematic review	11 RCTs	300; 13–28 y, 60%	LLLT 660–980 nm, 2–8 J/cm^2	Qualitative acceleration	65%; wavelength, duration	High	Moderate	LLLT accelerates OTM
14	Nayyer (2022) [3]	SR + meta-analysis	6 RCTs	240; 12–30 y, 58%	PBM 780–810 nm, 4–8 J/cm^2	Reduced OIIRR MD = 0.08 mm	88%; dosimetry	High	Moderate	PBM reduces orthodontically induced root resorption
15	Michelogiannakis (2019) [4]	Systematic review	8 RCTs	250; 14–28 y, 60%	LLLT 600–1200 nm, < 10 J/cm^2	Limited evidence of OIIRR reduction	NR	Critically Low	Low	Weak protective effect against OIIRR
16	Tini (2024) [12]	SR + meta-analysis	6 RCTs	200; 12–30 y, 55%	Vitamin D 1–50 µg/day or local	OTM acceleration (pooled effect positive)	NR	High	Moderate	Suggested possible acceleration, but evidence remained limited
17	Sagar (2024) [13]	Systematic review	5 RCTs	150; 15–25 y, 60%	1,25-dihydroxycholecalciferol local/oral	Acceleration reported	NR	Critically Low	Low	Promising but limited evidence
18	Al-Attar (2021) [14]	SR + meta-analysis	4 RCTs	120; 13–29 y, 62%	Calcitriol local/systemic	Variable impact; local > systemic	Moderate	Moderate	Low	Inconsistent results
19	Ferrillo (2024) [15]	Systematic review	10 RCTs	450; 12–34 y, 58%	Vitamin D local injections	Increased OTM, reduced OIIRR	High	High	Low	Local vitamin D injections promising
20	Gonçalves (2025) [18]	Systematic review	76 RCTs	Mixed adults/young	PBM & Vitamin D	PBM 25–59%, Vitamin D 10–51% acceleration	High	Moderate	Moderate	Non-surgical methods (including PBM & vitamin D) effective and promising

The conclusions in the final column are reproduced or paraphrased from the individual source SRMAs and reflect only the interpretation of those reviews. They do not represent the overall umbrella-level judgment of the present review

summarized narratively; we performed no independent assessments of these elements.

Results

The comprehensive search process identified 20 systematic reviews and meta-analyses (SRMAs). These reviews synthesized a large body of human orthodontic trials involving approximately 250 primary studies and several thousand participants overall. Quality assessments using AMSTAR-2 classified eight SRMAs as high-confidence, nine as moderate-confidence, and three as critically low-confidence (Table 3; Fig. 2). The ROBIS ratings indicated a low risk of bias in 12 SRMAs and an unclear risk of bias in eight (Table 4; Fig. 3).

Primary outcome: orthodontic tooth movement (OTM)

Because the source SRMAs used heterogeneous outcome metrics (mm displacement, mm/month, percentage acceleration, or phase-duration reduction) and differing follow-up intervals, the values reported below represent within-review signals only and are not directly comparable across reviews. Sixteen of the 20 included SRMAs evaluated LLLT/PBM, and four evaluated vitamin D. Low-certainty evidence from these reviews suggests that LLLT/PBM may be associated with some degree of OTM acceleration during various treatment phases. Individual high-confidence SRMAs reported additional OTM ranging from approximately 0.19 mm to 1.53 mm (or 24–59% faster movement) over varying observation intervals and treatment phases; however, because of differences in units, follow-up duration, and orthodontic mechanics, these estimates cannot be directly compared or pooled [8, 21, 41–43]. Several high- and moderate-confidence reviews also noted clinically relevant reductions in alignment time of approximately 25–30 days and overall treatment time reductions of 14–26% in certain phases [18, 43, 44]. Qualitative summaries in other reviews further supported a directional benefit for LLLT/PBM, although the magnitude varied depending on the wavelength (typically 660–980 nm), energy density (4–10 J/cm^2), and application frequency. Several reviews [7, 9–11, 19, 20] described acceleration as "possible" or "observed" but emphasized the high heterogeneity arising from differences in dosimetry, orthodontic mechanics, and patient age.

For vitamin D supplementation, the evidence remained at very low certainty and indicated no consistent association with OTM acceleration. Some reviews have suggested potential benefits of local administration [14, 15], while others have reported mixed findings across systemic and local routes [12, 13]. The overall body of evidence for vitamin D was limited by the small number of primary studies and marked variability in dosing, timing, and delivery methods. A study conducted by Gonçalves et al. [18] included vitamin D within a broader non-surgical adjunct analysis and reported acceleration estimates of 10–51%, but this was not consistent across the vitamin D-specific SRMAs.

Secondary outcomes

Low-certainty evidence suggests potential minor reductions in orthodontically induced inflammatory root resorption (OIIRR) associated with LLLT/PBM. The reported mean differences in root length loss ranged from approximately 0.08–0.13 mm in the included SRMAs, with effects primarily observed in the apical and middle thirds of the root on radiographic or CBCT imaging [3, 4]. The high-confidence review by Nayyer et al. [3] noted that these reductions were modest and varied according to the dosimetry and treatment phase. Limited and inconsistent evidence was reported for vitamin D on OIIRR, with some isolated signals for local injections, but insufficient for any clinical recommendation.

For pain, LLLT/PBM demonstrated possible analgesic effects, particularly for spontaneous and chewing pain, within the first 24–72 h after appliance activation. The standardized mean differences ranged from approximately −0.68 to −0.72 in pain-focused SRMAs [2] and related reviews. These short-term benefits were described as clinically relevant on visual analog scales, although longer-term pain trajectories were rarely assessed. Vitamin D was not evaluated for pain outcomes in the included SRMAs.

Evidence for bone biomarker modulation remains limited. Vitamin D-focused reviews, particularly Ferrillo et al. [15], suggested possible effects on the RANKL/OPG pathway, but the clinical significance remains uncertain.

Summary of findings

The certainty of evidence for all outcomes was rated as low or very low (Table 5). The certainty ratings in Table 5 represent umbrella-level judgments across the body of evidence and may differ from the certainty ratings reported within individual SRMAs in Table 2.

Discussion

Principal findings and interpretation

This umbrella review summarizes the findings of 20 systematic reviews and meta-analyses examining the adjunctive use of low-level laser therapy/photobiomodulation (LLLT/PBM) and vitamin D during fixed-appliance orthodontic treatment. Taken as a whole, the current evidence offers a modest but uncertain signal of benefit for LLLT/PBM across several outcomes, whereas the data for vitamin D continue to be limited and inconsistent. Although many of the included SRMAs reported directionally favorable results for LLLT/PBM, the overall certainty of evidence remains low to very low. This is largely

Table 3 AMSTAR-2 domains and overall confidence

No	Author (Year)	PICO	Protocol & Registration	Review of Search	Study Selection	Data Extraction	List of Excluded Studies	Description of Included Studies	RoB in Individual Studies	Sources of Funding	RoB in Synthesis	Heterogeneity	Publication Bias	Conclusions Appropriate	Overall Confidence
1	Domínguez Camacho (2020) [7]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
2	Jedliński (2020) [8]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
3	Ge (2014)	Yes	Partial	Yes	Yes	Partial	No	Yes	Partial	No	Partial	Yes	No	Yes	Moderate
4	Bakdach & Hadad (2020) [10]	Yes	No	Partial	Partial	Partial	No	Yes	Partial	No	Partial	Partial	No	Partial	Critically Low
5	de Almeida (2016) [11]	Yes	Yes	Yes	Partial	Partial	No	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
6	Deana (2017) [2]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
7	Imani (2018) [41]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
8	El-Angbawi (2023) [42]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
9	Zheng (2023) [43]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
10	Huang (2023) [44]	Yes	Yes	Yes	Partial	Partial	No	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
11	AlShahrania (2019)	Yes	Partial	Yes	Yes	Partial	No	Yes	Partial	No	Partial	Yes	No	Yes	Moderate

Table 3 (continued)

No	Author (Year)	PICO	Protocol & Registration	Review of Search	Study Selection	Data Extraction	List of Excluded Studies	Description of Included Studies	RoB in Individual Studies	Sources of Funding	RoB in Synthesis	Heterogeneity	Publication Bias	Conclusions Appropriate	Overall Confidence
12	Cronshaw (2019) [20]	Yes	Partial	Yes	Yes	Partial	No	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
13	Elgadi (2023) [21]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
14	Nayyer (2022) [3]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
15	Michelogiannakis (2019) [4]	Yes	Partial	Partial	Partial	Partial	No	Partial	Partial	Partial	Partial	Partial	No	Partial	Critically Low
16	Tini (2024) [12]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
17	Sagar (2024) [13]	Yes	Partial	Partial	Partial	Partial	No	Partial	Partial	Partial	Partial	Partial	No	Partial	Critically Low
18	Al-Attar (2021) [14]	Yes	Yes	Yes	Yes	Partial	No	Yes	Partial	Yes	Partial	Yes	No	Yes	Moderate
19	Ferrillo (2024) [15]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
20	Gonçalves (2025) [18]	Yes	Yes	Yes	Yes	Partial	No	Yes	Partial	Yes	Partial	Yes	Partial	Yes	Moderate

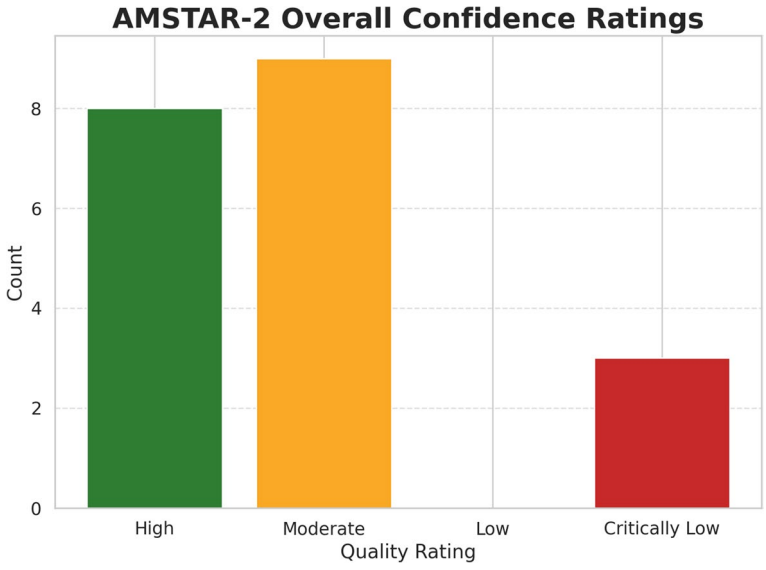


Fig. 2 AMSTAR-2 overall quality distribution

Table 4 ROBIS per domain and overall

No	Author (Year)	Domain 1 Study eligibility criteria	Domain 2 Identification & selection	Domain 3 Data collection & study appraisal	Domain 4 Synthesis & findings	Overall Risk of Bias
1	Domínguez Camacho (2020) [7]	Low	Low	Low	Unclear	Low
2	Jedliński (2020) [8]	Low	Low	Low	Low	Low
3	Ge (2014)	Low	Unclear	Low	Unclear	Unclear
4	Bakdach & Hadad (2020) [10]	Unclear	Unclear	High	Unclear	Unclear
5	de Almeida (2016) [11]	Low	Unclear	Unclear	Low	Unclear
6	Deana (2017) [2]	Low	Low	Unclear	Low	Low
7	Imani (2018) [41]	Low	Low	Low	Low	Low
8	El-Angbawi (2023) [42]	Low	Low	Low	Low	Low
9	Zheng (2023) [43]	Low	Low	Low	Low	Low
10	Huang (2023) [44]	Low	Unclear	Unclear	Low	Unclear
11	AlShahrania (2019)	Low	Unclear	Low	Unclear	Unclear
12	Cronshaw (2019) [20]	Low	Unclear	Low	Low	Low
13	Elgadi (2023) [21]	Low	Low	Low	Low	Low
14	Nayyer (2022) [3]	Low	Low	Low	Low	Low
15	Michelogiannakis (2019) [4]	Unclear	Unclear	Unclear	Unclear	Unclear
16	Tini (2024) [12]	Low	Low	Low	Low	Low
17	Sagar (2024) [13]	Unclear	Unclear	Unclear	Unclear	Unclear
18	Al-Attar (2021) [14]	Low	Low	Unclear	Low	Low
19	Ferrillo (2024) [15]	Low	Low	Low	Low	Low
20	Gonçalves (2025) [18]	Low	Unclear	Low	Low	Unclear

due to marked heterogeneity, differences in intervention protocols, and important limitations within the primary studies. These factors reduce confidence in the reproducibility and clinical generalizability of the reported effects [8, 12–15, 21, 41–43].

For orthodontic tooth movement (OTM), evidence suggests that LLLT/PBM may accelerate tooth displacement in certain treatment phases, most notably during canine retraction and initial alignment. However, the magnitude of this effect varies considerably across studies and appears to depend heavily on wavelength, energy density, application frequency, specific orthodontic mechanics employed, and length of follow-up [7–11, 19–21, 41–44]. This pattern implies that the intervention may exert a possible biological influence on bone remodelling; however, the lack of standardization still prevents confident translation into everyday clinical practice. Even when statistically significant improvements were noted

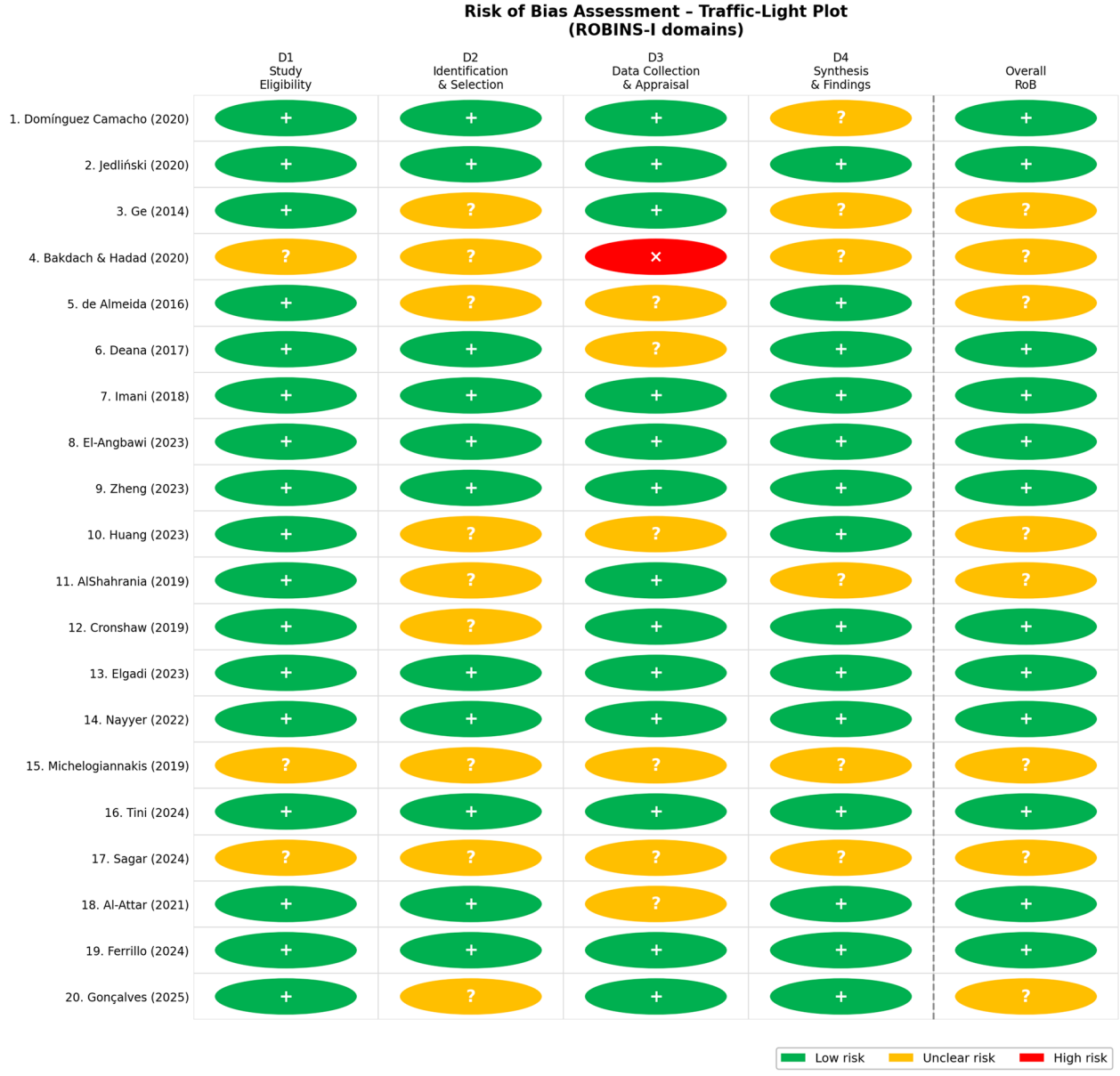


Fig. 3 ROBIS risk of bias overall distribution

in individual meta-analyses, the practical clinical gain appeared relatively small. In real-world terms, such acceleration may translate into only a modest reduction in overall treatment time, and any potential advantage must be carefully balanced against added chairside time, protocol complexity, patient acceptability, and value in clinical practice [5, 6].

The secondary outcomes followed a similar pattern. With regard to pain, LLLT/PBM seems to provide a short-term analgesic effect in the first 24–72 h after appliance activation, an outcome that may hold meaningful value from the patient's perspective [2, 42, 45]. For orthodontically induced inflammatory root resorption (OIIRR), the data suggest a modest protective influence, although the clinical relevance of the small reduction observed remains uncertain [3, 4]. Changes in bone biomarkers, particularly along the RANKL/OPG pathway, also appeared directionally favorable in selected reviews, but the limited number of contributing studies and the surrogate nature of these markers make it difficult to draw firm clinical conclusions [15]. Collectively, these results support a degree of biological plausibility for LLLT/PBM; however, they do not constitute a sufficiently robust evidence base to justify routine clinical adoption.

Vitamin D, despite a plausible mechanistic rationale through its actions on the vitamin D receptor and subsequent effects on bone remodelling, failed to show consistent clinical benefit across the eligible SRMAs

Table 5 Summary of Findings (SoF) Table

Outcome	Intervention	Anticipated absolute effects (range reported across SRMAs)	Certainty of the evidence (GRADE)	Number of participants (number of SRMAs)	Comments
Orthodontic Tooth Movement (OTM)	LLLT/PBM	Additional displacement 0.19–1.53 mm over varying periods; mixed acceleration reported	Low ⊕⊕◯◯	~6,000 (16)	High heterogeneity; clinical significance uncertain
OTM	Vitamin D	Inconsistent findings; no reliable association in most reviews	Very low ⊕◯◯◯	~1,000 (4)	Sparse data; high variability; local administration explored in subgroups only
Orthodontically Induced Inflammatory Root Resorption (OIIRR)	LLLT/PBM	Minor reductions in root length loss (~0.08–0.13 mm in some reviews)	Very low ⊕◯◯◯	600 (3)	Limited number of SRMAs; influenced by dosimetry
OIIRR	Vitamin D	Limited/inconsistent data; isolated signals for local injections (Ferrillo 2024) but insufficient for recommendation	Very low ⊕◯◯◯	~450 (1)	Very sparse data
Pain (24–72 h post-activation)	LLLT/PBM	Possible reduction in pain intensity (SMD ~ −0.70)	Low ⊕⊕◯◯	1,000 (2)	Short-term effect only
Pain	Vitamin D	No data available	—	N/A	Not evaluated
Bone Biomarkers (e.g., RANKL/OPG ratio)	LLLT/PBM	Sparse data; possible modulation suggested	Low ⊕⊕◯◯	400 (2)	Limited evidence
Bone Biomarkers	Vitamin D	Limited/inconsistent data; isolated signals for modulation but insufficient for recommendation	Low ⊕⊕◯◯	400 (2)	Limited number of SRMAs

[12–15]. Some reviews have indicated that local delivery may be more promising than systemic supplementation; however, these subgroup observations have not been adequately replicated and remain constrained by small sample sizes and considerable protocol heterogeneity [12–15]. VDR-related interindividual variation remains biologically plausible, but it has not yet been integrated meaningfully into the current clinical review evidence base [15, 25]. Consequently, the present body of evidence for vitamin D should still be viewed as exploratory, with isolated signals for local administration that remain insufficient for clinical recommendations.

Comparison with existing evidence

The present findings are broadly consistent with those of the orthodontic literature. Systematic reviews published between 2014 and 2019 were generally cautious yet optimistic about the potential of LLLT/PBM to accelerate OTM, although their conclusions were tempered by relatively few primary studies and variable methodological quality [7–11, 19, 20]. The Cochrane review by El-Angbawi et al. remains particularly influential, as it offers a more conservative estimate of benefit and rightly emphasizes the ongoing clinical uncertainty despite some positive signals [42]. Later SRMAs that focused on specific treatment phases especially canine retraction and alignment tended to report somewhat larger effect sizes, raising the possibility that any advantage of LLLT/PBM may be phase-dependent rather than consistent throughout the entire course of treatment [8, 41, 43, 44].

The pain-related findings are also consistent with the wider evidence. The short-term analgesic benefit documented in the pain-focused SRMAs aligns well with the recent meta-analysis of meta-analyses, which demonstrated pain reduction with both LLLT and ibuprofen during orthodontic treatment, while again highlighting substantial heterogeneity in the LLLT analyses [45]. This reinforces the idea that pain relief may represent one of the more reproducible effects of LLLT/PBM, although questions regarding the optimal dose, timing, and device characteristics have yet to be fully resolved.

More recent high-level evidence, including the PBM-focused umbrella review by Jiménez-Peña et al. [39] and the scoping review by Salha et al. [46], arrived at similarly cautious conclusions. Although these studies were not formally included in the present synthesis because of our predefined eligibility criteria, they provide valuable contextual information. Both highlighted possible benefit signals while underscoring persistent heterogeneity, limited standardization, and inconsistent review quality [39, 46]. In essence, the current umbrella review consolidates recent literature and places it in a broader clinical context. Earlier reviews on surgical and non-surgical acceleration methods [16, 17] similarly noted the need for better standardization. Other systematic reviews have also examined the combination of self-ligating brackets with various acceleration methods [47]. This perspective is further supported by the large 2025 systematic review by Gonçalves et al. [18], who described promising but variable outcomes for non-surgical acceleration methods.

The evidence trajectory has remained comparatively weak. Despite clear mechanistic plausibility and occasional favorable subgroup findings, the four dedicated SRMAs have not resolved the core inconsistencies in the literature [12–15]. Local administration has appeared more encouraging than systemic supplementation in some analyses; however, the overall evidence base is still too small and heterogeneous to inspire confidence in this distinction [12–15]. Overall, LLLT/PBM has grown quantitatively without a comparable qualitative advance in certainty, whereas vitamin D continues to reside in an exploratory phase of investigation.

Research gaps and future directions

This review highlights several enduring limitations of the existing evidence base. First, outcome standardization remains inadequate, particularly for orthodontic tooth movement. The source SRMAs employed different units, observation intervals, and phase-specific definitions, which complicate meaningful comparisons and restrict the potential for robust synthesis [18, 43, 44]. Future investigators should adopt standardized, clinically meaningful outcome measures, together with clearly defined time points and consistent reporting frameworks [29, 31, 48].

Second, the literature has paid remarkably little attention to the biological and contextual sources of variability. Ethnic, geographic, and genetic differences are rarely examined, even though responsiveness to both vitamin D and possibly LLLT/PBM may be influenced by genetic variants, such as VDR polymorphisms [15, 25]. None of the included SRMAs could incorporate these factors meaningfully, primarily because the underlying trials did not collect or report the necessary data. This represents a significant limitation to external validity and should be addressed in future multicenter studies that include more diverse populations and prospectively planned subgroup analyses.

Third, long-term outcomes have been neglected almost entirely. The available evidence concentrates on short-term tooth movement rates, pain scores, and limited biomarker changes, while important clinical endpoints, such as post-treatment stability, relapse rates, and the progression of root resorption over time, are largely absent. This leaves a major gap in our understanding: even if an adjunct modestly speeds up movement during active treatment, it remains unknown whether that benefit persists without compromising long-term occlusal stability or root health [3, 4].

Fourth, patient-centered outcomes are seriously underreported. Compliance, treatment experience, oral health-related quality of life, patient satisfaction, and cost-effectiveness are directly relevant to real-world applications; however, these measures were addressed in only a small minority of the primary studies [5, 6]. This is especially important because the practical value of any modest acceleration effect ultimately depends not only on biological efficacy but also on feasibility, affordability, and patient acceptability.

Although several individual SRMAs reported directionally favorable findings, umbrella-level interpretation must be more conservative because the totality of evidence is affected by substantial heterogeneity, inconsistency, imprecision, and variable review quality. Finally, the evidence base is constrained by many small and underpowered primary trials. A substantial proportion of the RCTs included in the source SRMAs enrolled modest numbers of participants, contributing to imprecision, wide confidence intervals, and unstable pooled estimates [2–4, 8–15, 41–44]. Therefore, the field appears to have reached a point at which further progress will depend less on simply conducting more small heterogeneous studies and more on performing rigorously designed, adequately powered, and well-standardized trials. The future research should prioritize multicenter RCTs with harmonized intervention protocols, transparent reporting standards, extended follow-up periods, validated patient-reported outcome measures, and, where appropriate, factorial designs that explore combined or stratified intervention approaches.

Strengths and limitations

A key strength of this umbrella review is its strict focus on human clinical evidence evaluated using well-established methodological frameworks, including AMSTAR-2 [27], ROBIS [30], GRADE [28], and umbrella-review guidance [29]. The review considered overlap narratively in line with umbrella-review methodology and maintained a clear clinical emphasis on the outcomes most relevant to orthodontists: OTM, pain, OIIRR, and bone biomarkers [49]. In addition, the final synthesis deliberately adopted a conservative narrative approach, which is appropriate given the high degree of heterogeneity across reviews and helps avoid the risk of overinterpreting unstable estimates across reviews [29, 48].

Nevertheless, several limitations of this study should be acknowledged. First, umbrella reviews are inherently dependent on the quality and completeness of the included SRMAs and their underlying primary studies. In the present case, heterogeneity in dosimetry, treatment mechanics, outcome definitions, and follow-up duration was substantial [2–4, 7–15, 19–21, 41–44]. Second, the certainty of the evidence was frequently downgraded owing to inconsistency, imprecision, and risk of bias within the primary literature [27, 28, 50]. Third, certain outcomes, especially those related to vitamin D, OIIRR, and bone biomarkers, were informed by relatively few SRMAs [3, 4, 12–15]. Fourth, the majority of

primary trials involved adolescents or young adults, which restricts the generalizability of the results to broader orthodontic populations. Finally, the absence of long-term follow-up data limits any confident assessment of treatment durability, stability, or safety beyond the short-term follow-up phase.

Clinical and research implications

From a clinical perspective, the present evidence does not support the routine incorporation of LLLT/PBM or vitamin D as adjuncts in orthodontic practice. For LLLT/PBM, the most consistent signals relate to modest short-term acceleration and pain relief; however, these benefits are not yet backed by evidence that is sufficiently consistent, standardized, or precise to warrant broad implementation [2–4, 8, 41–44]. For vitamin D, the data remain too limited and inconsistent to justify its clinical use outside research settings [12–15]. Therefore, any clinical interest in these modalities should remain cautious and firmly grounded in the available evidence.

From a research perspective, the priority is no longer simply to conduct more studies but to conduct better studies. The field would benefit most from standardized intervention protocols, clearer reporting of key parameters, consensus-driven outcome measures, extended follow-up, and greater integration of biological and patient-centered variables [5, 6, 29, 31, 48]. Only through such advances will it become possible to establish whether these adjuncts can deliver meaningful clinical benefits rather than effects that are statistically detectable but of uncertain practical importance.

Research recommendations

Future research should prioritize adequately powered multicenter RCTs with standardized protocols, longer follow-ups, biomarker assessments, and economic evaluations.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12903-026-09095-8.

Supplementary Material 1.

Acknowledgements
None.

Authors' contributions
S.A.A.B: Conceptualization, Methodology, Investigation, Writing—Original Draft, Visualization. A.H.R: Formal Analysis, Validation, Writing—Review & Editing. R.H: Resources, Supervision, Writing, reviewing, and editing. All authors have read and approved the final manuscript.

Funding
This study did not receive any external funding.

Data availability
All data generated or analysed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate
Not applicable, as this was a review of published literature.

Competing interests
The authors declare no competing interests.

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Received: 30 November 2025 / Accepted: 23 June 2026
Published online: 16 July 2026

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